LESSON 5.10

Volume of Rectangular Prisms

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can find the volume of a rectangular prism, including ones with fractional edge lengths, using base area × height.

WHY THIS MATTERS Your class is building a time capsule to bury on school grounds. The capsule is a rectangular prism that is 2 feet long, 1.5 feet wide, and 1 foot tall. You need to figure out how much stuff can fit inside — that means calculating the volume!

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Volume *Spanish: Volumen*	How much space is inside a solid shape.	A box 2 × 1.5 × 1 = 3 ft³ — it holds 3 cubic feet of stuff
Rectangular prism *Spanish: Prisma rectangular*	A solid box shape with six flat rectangle sides.	A tissue box, a fish tank, a brick — all rectangular prisms
Cubic units *Spanish: Unidades cúbicas*	The units used to measure space inside, like cubic inches.	1 ft³ = a cube that is 1 foot on every edge
Dimensions *Spanish: Dimensiones*	How long, how wide, and how tall a shape is.	A box with dimensions 4 × 3 × 2 means l = 4, w = 3, h = 2
Base area *Spanish: Área de la base*	The area of the bottom of a solid. Volume = base area × height.	If the base is 6 × 4 = 24 cm² and h = 3, then V = 24 × 3 = 72 cm³

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

My capsule is 2 ft long, 1.5 ft wide, and 1 ft tall. I write V = length × width × height.

1. I put in the numbers: V = 2 × 1.5 × 1.
2. First I multiply 2 × 1.5 = 3.
3. Then 3 × 1 = 3.
4. So the volume is 3 ft³. A fractional edge like 1.5 did not change my steps at all.

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

Now try a box that is 4 long, 2 wide, and 0.5 tall. Let's do it the base-area way. What is the base, 4 × 2?

1. _____
2. _____
3. _____
4. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- Capsules A and B both have V = _____ because _____ × _____ × _____ = _____ × _____ × _____. I'd choose _____ because _____.

Word bank Volume | Rectangular prism | Cubic units | Dimensions | Base area | 3 | volume | same | different | dimensions | shape | length

WATCH OUT!

A common mistake

A common mistake in Volume of Rectangular Prisms is rounding a fractional edge length to the nearest whole number before multiplying, instead of using the exact fraction or decimal. For a box that is 2 ft × 1.5 ft × 1 ft, a student might round 1.5 down to 1 and compute 2 × 1 × 1 = 2 ft³, but the correct volume uses the exact edge length: 2 × 1.5 × 1 = 3 ft³

LESSON 5.10

Volume of Rectangular Prisms

Level 0 · Most support · every hint printed

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Complete the table. Show how you found each value.**

Find the volume of each rectangular prism.

Length	Width	Height	Volume
8 in	5 in	2 in	
4 cm	4 cm	4 cm	
10 ft	3 ft	6 ft	

STEPS TO TRY
1. Each row lists a length, width, and height. The blank Volume cell needs a number with cubic units.
2. For every row use V = l × w × h: multiply the first two numbers, then multiply by the third.
3. Row 1: multiply 8 × 5 = 40 first, then multiply by the height, 2. Include in³ in your answer.

SHOW YOUR WORK

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

What is the volume of a rectangular prism with l = 4 in, w = 3 in, h = 5 in?

(A) 12 in³
(B) 24 in³
(C) 60 in³
(D) 60 in²

STEPS TO TRY

1. Spot the three edges: l = 4 in, w = 3 in, h = 5 in. Volume fills the whole box, so all three matter.
2. Use V = l × w × h and remember the answer needs cubic units (in³), not square units.
3. Multiply 4 × 3 = 12 for the base, then multiply that result by the height, 5 inches.

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

1. **Name it** *Which measure is asked?*
2. **Write the formula** *Before any numbers.*
3. **Substitute** *Match each letter.*
4. **Unit** *units, sq units, or cubic?*

3 **Circle the letter of the best answer. Show how you know.**

A box has a volume of 36 cm³. Its length is 6 cm and width is 3 cm. What is the height?

(A) 2 cm
(B) 3 cm
(C) 6 cm
(D) 12 cm

STEPS TO TRY

1. You're given the volume (36 cm³) plus two edges (6 cm and 3 cm). The height is what's missing.
2. Work backward: divide the volume by the base area (length × width) to uncover the height.
3. Find the base area first: 6 × 3 = 18 cm². Then ask what number times 18 gives 36.

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

1. **Name it** *Which measure is asked?*
2. **Write the formula** *Before any numbers.*
3. **Substitute** *Match each letter.*
4. **Unit** *units, sq units, or cubic?*

4 **Circle the letter of the best answer. Show how you know.**

How is volume different from area?

(A) Volume uses cubic units; area uses square units

(B) Volume uses square units; area uses cubic units

(C) Volume only applies to cubes; area applies to all shapes

(D) There is no difference

STEPS TO TRY

1. Think dimensions: area covers a flat 2D surface, while volume fills a 3D space.
2. Match each measurement to its unit: flat surfaces get square units (^{2}), filled space gets cubic units (^{3}).
3. Picture a rug (flat) versus a fish tank (filled). Decide which is area and which is volume, then check each choice.

SHOW YOUR WORK

EXPLAIN YOUR THINKING

Capsules A and B have the same volume (3 ft^3) but different dimensions. Why? Which design would you choose and why?

SENTENCE STARTER Capsules A and B both have V = ___ because ___ × ___ × ___ = ___ × ___ × ___. I'd choose ___ because ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help